

Crypto Intraday Trading System

Jev + Multi-Agent LLM

Paper-only architecture for BTCUSDT perpetual futures · isolated 3x · deterministic risk control

STATUS M1 foundation	OWNER Julius	LAST UPDATED September 22, 2026
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Reviewers	Risk, engineering, and operations reviewers before implementation
Related docs	docs/architecture.md · docs/evaluation.md · docs/btc-paper-runbook.md
Scope	Standalone BTCUSDT perpetual paper system; the daily Donchian bot remains unchanged.

Design artifact · not an authorization for live trading

1. Abstract

This document proposes a 24/7 crypto intraday paper-trading system for Binance USD-M BTCUSDT perpetual futures. It combines a slow LLM analysis layer, which produces market theses and bounded rule candidates every 60 minutes, with a fast Jev decision layer that evaluates synchronized market features every five seconds. Jev is the primary source of directional decisions, while deterministic code owns validation, position sizing, margin limits, liquidation protection, idempotency, and all hard risk controls.

The system uses isolated margin with leverage fixed at 3x, one-way positions, and at most two entry tranches. Leverage is a buying-power ceiling, not a multiplier for the risk budget. Each tranche risks at most 0.25% of account equity; the complete position risks at most 0.50%, uses at most 50% notional exposure, and reserves no more than 20% of equity as initial margin. A daily loss limit of 1.5% and a high-water-mark drawdown kill switch of 8% remain immutable.

The first release is paper only. It has no exchange trading credentials, no endpoint that can place a live order, and no migration path that silently converts paper orders to live orders. A later live-trading proposal would require a separate design review, new credentials and permissions, exchange reconciliation, and an independent launch gate.

2. Goals and Non-Goals

Goals	Non-goals
Produce a fresh, auditable Jev decision every five seconds when required inputs are healthy.	Execute real-money orders or hold an API key with trading permission.
Simulate perpetual-futures fills, fees, funding, margin, and liquidation risk with reproducible accounting.	Support cross margin, hedge mode, leverage above 3x, options, or spot execution.
Let an LLM propose declarative rules without granting it control of hard risk limits.	Let generated text or code execute directly in the trading process.
Promote challenger rules only after replay and a 14-day/30-trade shadow evaluation.	Optimize a challenger on the same evaluation window used to approve it.
Provide a private dashboard, Telegram operations channel, audit trail, and safe operator controls.	Build a public multi-user SaaS dashboard or a complex SPA.
Run continuously on one Linux VPS using Docker Compose and a recoverable SQLite data store.	Provide active-active availability or multi-region operation in v1.

3. Background and Problem Statement

The existing repository contains a deterministic, daily BTC paper-trading campaign based on closed candles and a promoted Donchian strategy. It is intentionally narrow, reproducible, and conservative. An intraday system that changes rules through an LLM, uses order-book and derivatives data, and calls a model every five seconds has a different failure surface and must not mutate the promoted campaign or share its paper account.

The original intraday prompt leaves several safety-critical details underspecified:

- A mutable `rules_current.json` can change behavior mid-decision and cannot provide a complete replay trail.
- A 30-minute champion/challenger contest measures noise rather than durable strategy quality.
- Repeated REST requests every five seconds produce temporally inconsistent features and unnecessary rate-limit pressure.
- A single confidence threshold does not protect against stale data, malformed model output, over-sizing, liquidation proximity, or repeated orders.
- Editing `.env` from a remote dashboard creates a secret-exposure and configuration- integrity risk.
- Cost assumptions are not credible until actual request counts and input tokens are measured. A five-second loop alone can make 17,280 Jev calls per day.

The governing invariant is therefore:

Models may recommend exposure, but only deterministic, versioned, and replayable policy may authorize a paper fill.

4. Proposed Architecture

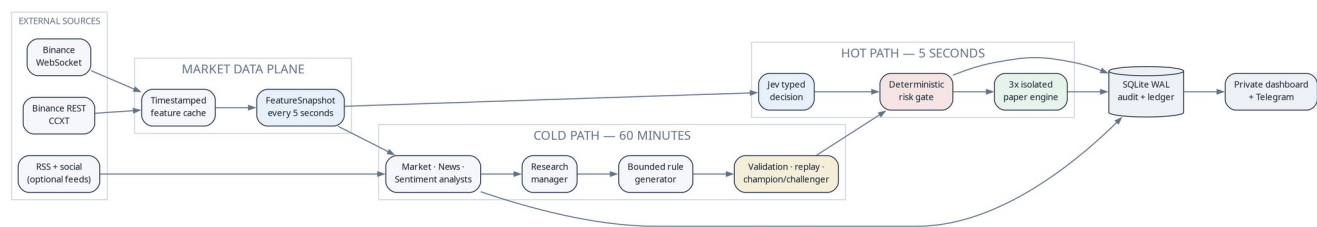


Figure 1. Proposed system architecture and trust boundaries.

Core components

Component	Responsibility	Primary state	Failure behavior
Market data plane	Maintain WebSocket streams, poll slower derivatives data, calculate indicators, and emit synchronized snapshots.	In-memory cache plus persisted snapshots	Mark affected fields stale; block new entries when required fields exceed TTL.
LLM analysis layer	Produce evidence-backed analyst reports, a market thesis, and bounded rule candidates.	Versioned reports and candidates in SQLite	Keep the last valid thesis; do not create or activate a candidate.
Jev decision provider	Send one state snapshot with typed questions and normalize typed answers and confidence.	Immutable decision records	Timeout or invalid response becomes HOLD; no retry inside the same decision tick.
Deterministic risk gate	Validate freshness, rules, confidence, exposure, margin, liquidation buffer, daily loss, and drawdown.	Active rule/config versions and risk state	Fail closed for entries; risk-reducing exits remain available.
Paper execution engine	Simulate market fills, one-way positions, fees, funding, margin, mark-to-market, and liquidation safeguards.	Paper ledger, orders, fills, positions	Commit nothing unless order and ledger transition can be atomic.
Rule evaluation service	Replay candidates and run champion/challenger ledgers against identical inputs.	Candidate lifecycle and evaluation windows	Preserve champion; pause evaluation and surface missing coverage.
Operations plane	Serve private monitoring/control views and send deduplicated Telegram events.	Audit log and notification outbox	Trading remains safe if UI or Telegram is unavailable.

Deployment boundary

Docker Compose runs two application services on one Linux VPS:

- worker: ingestion, feature calculation, LLM schedules, Jev loop, risk gate, paper execution, evaluations, and notification outbox.
- web: read APIs, server-rendered dashboard, and narrowly scoped operator commands.

Both services mount one persistent volume. SQLite runs in WAL mode; the worker is the single writer for trading state. The web process submits control commands into an inbox table rather than mutating positions or rules directly. Nightly backups copy a consistent SQLite snapshot to storage outside the container volume.

Cross-venue market evidence

Binance remains the reference and simulated execution venue. Hyperliquid is an optional evidence source: BTC L2 book arrives over WebSocket while funding, open interest, mark, and oracle context refresh over public REST every 30 seconds. The worker normalizes funding to basis points per hour, open interest to USD, and book depth/imbalance inside 5/10/25 bps bands before binding the evidence into the immutable five-second snapshot.

The overlay defaults to shadow. Confirming evidence may recommend +0.5 entry quality; conflicting evidence recommends -1.0 and a 0.5 notional multiplier; verified venue stress recommends no new exposure. It never changes direction, confidence, leverage, stop-loss, or any hard-risk limit. Missing Hyperliquid evidence is neutral and leaves

Binance sizing unchanged. Activation requires a persisted 14-day promotion evaluation; changing configuration alone cannot bypass the gate. Lighter is deferred until the Hyperliquid pipeline completes its promotion evaluation and will reuse the same venue frame interface.

News-source policy

The allowlist is verified independently of model prompts. V1 enables the public RSS feeds from the SEC, CFTC, Federal Reserve, CoinDesk, Decrypt, and Cointelegraph. Binance announcements remain disabled until a separate signed, read-only announcement credential is approved. The Block is not sent to an LLM because its current terms prohibit automated AI/ML processing of its content. Wu Blockchain remains disabled until a stable, authorized machine-readable endpoint is verified. Disabled sources remain visible in the source catalog with a reason; the system never silently replaces them with scraping or an unofficial mirror.

5. Decision and Analysis Lifecycles

Five-second decision lifecycle

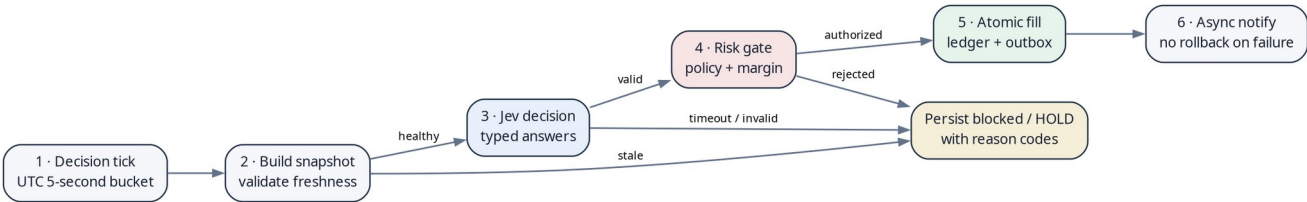


Figure 2. Five-second decision lifecycle and fail-closed paths.

- 1. The scheduler creates decision_tick_id from the UTC five-second bucket and symbol.
- 2. The feature builder reads the latest stream state, closed 1h indicators, funding, open interest, long/short ratio, and cached sentiment. It persists the exact input.
- 3. Required features must meet per-field TTLs. A stale required field produces a persisted blocked decision without calling Jev.
- 4. The Jev adapter calls OpenRouter using base URL https://openrouter.ai/api/v1 and pinned model typesafe/jev-1.13. It records latency, token usage, provider request ID, model ID, and the raw typed response.
- 5. The gate maps the response to a target exposure and evaluates it against the active immutable rule version and current hard-risk state.
- 6. An authorized exposure transition becomes one idempotent simulated market order. Order, fill, ledger, position, audit event, and notification-outbox row commit in a single transaction.
- 7. A notification worker sends outbox events and records delivery separately. Delivery failure never rolls back a fill.

Direction-to-exposure mapping

Jev direction	Flat account	Existing same-direction position	Existing opposite position
Strong Buy	Target +2 long tranches	Add up to +2	Close short; do not open long in the same tick
Buy	Target +1 long tranche	Add up to +1	Close short; do not open long in the same tick
Hold	Stay flat	Keep current exposure	Keep current exposure
Take Profit	No-op	Reduce one tranche only when unrealized P&L is positive	Reduce one tranche only when profitable
Sell	Target -1 short tranche	Add up to -1	Close long; do not open short in the same tick
Strong Sell	Target -2 short tranches	Add up to -2	Close long; do not open short in the same tick

Entry transitions have a 60-second cooldown. Closing, stop-loss, liquidation protection, daily stop, and kill-switch actions bypass the cooldown and are always reduce-only.

Hourly analysis and rule lifecycle

1. Market, news, and sentiment analysts independently create schema-valid reports with cited evidence, source times, and explicit missing-data fields.
2. The research manager consumes only those structured reports and produces a market thesis: bias, regimes, levels, risks, time horizon, evidence IDs, and uncertainty.
3. The rule generator may create a candidate only when the thesis materially changes and the 24-hour proposal cooldown has expired.
4. External content remains quoted data, never system instructions. The generated candidate is declarative JSON and cannot contain expressions, code, URLs, tools, or unknown keys.
5. The validator checks schema, allowlists, numeric bounds, and the immutable hard-risk envelope. Invalid candidates are rejected with structured reasons.
6. A valid candidate runs against the trailing 90 days with chronological inputs, closed-candle indicators, fees, funding, spread, and slippage.
7. At most one replay-passed challenger runs against the champion. Both receive the same snapshots but use independent ledgers and identical starting equity.
8. Evaluation lasts at least 14 calendar days and 30 closed challenger trades. New proposals queue without replacing the active challenger.
9. Promotion is an atomic registry transition. The former champion becomes the immediate rollback target. Rejected candidates enter the evaluation archive; promoted versions enter the hall of fame.

Promotion formula

For a common evaluation interval:

```
risk_adjusted_score = net_return_pct / max(max_drawdown_pct, 0.5)
```

A challenger is eligible only when all conditions hold:

- Net return after fees, funding, and slippage is positive.
- Required-data coverage is at least 99%.
- Account drawdown stays below 8% and no hard-risk invariant is violated.
- When champion score is positive, challenger score is at least 10% higher.
- When champion score is zero or negative, challenger score is greater than 0.25.

Promotion is deferred—not failed—when either the duration or trade-count requirement has not been met.

6. API and Data Contracts

All persisted contracts carry schema_version, a stable identifier, UTC timestamps, and provenance. Pydantic rejects unknown fields at model boundaries.

Core contracts

Contract	Required fields	Purpose and guarantees
FeatureSnapshot	snapshot_id, symbol, event_time, built_at, features, freshness, quality, checksum	Exact Jev/LLM input; immutable and content-addressed for replay.
AnalystReport	report_id, kind, as_of, evidence[], findings, uncertainty, model_ref, prompt_version	Structured analyst output; external evidence remains data.
MarketThesis	thesis_id, bias, regime, levels, risks, horizon, evidence_ids[]	Research manager synthesis; never authorizes a trade.
RuleCandidate	rule_id, parent_rule_id, thesis_id, parameters, constraints, content_hash	Declarative proposal validated against an allowlist and hard bounds.
JevDecision	decision_id, tick_id, snapshot_id, direction, direction_confidence, regime, toxic_flow, entry_quality, risk_level, model_ref	Typed decision and the probability/score used by the gate.
GateDecision	gate_id, decision_id, rule_id, risk_state_id, outcome, reason_codes[], target_exposure	Authoritative record of why exposure was authorized or rejected.

Contract	Required fields	Purpose and guarantees
PaperOrder	order_id, idempotency_key, side, quantity, reduce_only, created_at	Requested simulated market transition; no live-exchange fields.
PaperFill	fill_id, order_id, price, quantity, fee, slippage, filled_at	Immutable execution fact used by the ledger.
PositionSnapshot	side, quantity, entry_price, mark_price, notional, leverage, isolated_margin, maintenance_margin, liquidation_price, liquidation_buffer, funding, unrealized_pnl	Complete one-way perpetual position state.
RiskState	equity, high_water_mark, daily_pnl, drawdown, position_risk, margin_used, halt_state	Source of truth for deterministic authorization.
PromotionEvaluation	candidate_id, champion_id, window, coverage, trades, net_returns, drawdowns, scores, status, reasons[]	Reproducible promotion evidence.

Jev question contract

- direction: Choice of Strong Buy, Buy, Hold, Take Profit, Sell, or Strong Sell.
- regime: Choice of Trending Up, Trending Down, Sideways, or Volatile.
- toxic_flow: Noul probability that current microstructure indicates manipulative or adverse large-order flow.
- entry_quality: Score from 1 to 5 using a fixed legend.
- risk_level: Score with fixed labels Low, Medium, High, and Critical.

direction_confidence is the returned probability for the selected direction, not an average across questions. Until local calibration tests establish otherwise, toxic_flow is treated as a model score with a probability-shaped range rather than as a statistically proven market probability.

Gate order

The gate evaluates checks in this order and persists every failed reason:

1. Global and daily halt state.
2. Snapshot integrity, required fields, and freshness.
3. Model response validity and pinned version.
4. Direction confidence at least 0.85.
5. Entry quality at least 3.
6. Toxic-flow score at most 0.30.
7. Risk level Low or Medium for new exposure.
8. Active rule allowlists and session/regime filters.
9. Entry cooldown and idempotency.
10. Position, notional, margin, stop risk, and liquidation-buffer limits.

Risk-reducing actions skip entry-only checks but must remain reduce-only and valid for the current position.

Hard-risk policy

```

MARGIN_MODE=isolated
LEVERAGE=3
MAX_TRANCHES=2
MAX_TRANCHE_RISK_PCT=0.0025
MAX_POSITION_RISK_PCT=0.005
MAX_TRANCHE_NOTIONAL_PCT=0.25
MAX_POSITION_NOTIONAL_PCT=0.50
MAX_INITIAL_MARGIN_PCT=0.20
DAILY_LOSS_LIMIT_PCT=0.015
MAX_DRAWDOWN_PCT=0.08
MIN_STOP_DISTANCE_PCT=0.0075
MAX_STOP_DISTANCE_PCT=0.03
MIN_LIQUIDATION_BUFFER_PCT=0.15
EMERGENCY_LIQUIDATION_BUFFER_PCT=0.10
ENTRY_COOLDOWN_SECONDS=60

```

These values belong to a versioned deployment policy. The dashboard, LLM, Jev, rule candidates, and runtime .env editor cannot change them.

Position sizing

For each tranche:

```
risk_limited_notional = (equity * 0.0025) / stop_distance_pct
tranche_notional = min(equity * 0.25, risk_limited_notional)
position_notional = min(sum(tranches), equity * 0.50)
initial_margin = position_notional / 3
```

The gate rejects quantities below Binance symbol minimums after tick/step-size rounding. It also rejects additions that would make initial margin exceed 20% of equity. Liquidation price uses the current Binance maintenance-margin tier and mark price, not a fixed percentage approximation.

Dashboard interface

Read endpoints expose status, snapshots, decisions, gate reasons, paper positions, ledger, rules, reports, evaluations, costs, and service health. The only mutation commands are:

- Pause new entries.
- Resume entries after an explicit confirmation.
- Flatten the paper position with a reduce-only command.
- Roll back to the immediately preceding champion.
- Enable or disable non-critical Telegram notifications.

Commands are authenticated, CSRF-protected, idempotent, and appended to the audit log. The dashboard never reads or writes secret values.

7. Consistency, Idempotency, and Replay

Scenario	Expected behavior	Mechanism
The same five-second tick runs twice	At most one order/fill transition exists; both attempts resolve to the same decision record.	Unique decision_tick_id and order idempotency key.
Jev times out or returns invalid data	Persist a HOLD fallback and provider error; do not retry within the tick.	Bounded timeout and strict response validation.
SQLite transaction fails	No partial order, fill, position, ledger, or outbox state is visible.	One atomic transaction with rollback.
Telegram send fails	Trading record remains committed; notification retries with backoff and deduplication.	Transactional outbox and delivery key.
Configuration changes during a decision	The versions captured when the decision started remain authoritative.	Immutable config/rule IDs on every decision.
Worker restarts with an open position	Rebuild state from ledger, verify the latest position snapshot, then resume decision ticks.	Event/ledger reconciliation before scheduler readiness.
Market stream reconnects	New entries remain blocked until snapshots are fresh and sequence gaps are resolved.	Stream sequence tracking and readiness gate.
Replay is requested	Reuse stored snapshots, rule/config versions, model outputs, and execution assumptions.	Immutable provenance and content hashes.

Internal times are UTC. Event time determines market ordering; receive time measures latency. The system never silently substitutes receive time for a missing event time.

8. Security and Privacy Considerations

- The VPS exposes no public dashboard port. Operators connect through an SSH tunnel or private VPN; FastAPI binds to localhost by default.

- API credentials live in a root-readable secrets file or container secret, outside the repository and database. Logs redact authorization headers and secret-shaped fields.
- V1 does not store a Binance trading key. Public market-data access is sufficient for the paper venue.
- RSS, Reddit, X, and all article bodies are untrusted text. They are length-limited, tagged by source, passed only in data fields, and prevented from selecting tools, schemas, models, URLs, or system prompts.
- Generated rules use strict JSON schemas with unknown fields forbidden. No expression evaluation, dynamic imports, subprocesses, or generated Python are supported.
- Operator commands use session authentication, CSRF protection, confirmation for flatten/rollback, and immutable audit records.
- Raw social/news content has a 30-day retention default; normalized evidence and hashes are retained with the evaluation record. Trading/audit records are retained for the entire paper campaign.
- Backups are encrypted in transit and at rest, tested monthly, and never include environment-secret files.

9. Operational Readiness

Signal	SLO or alert	Launch gate
Decision-loop availability	At least 99% of scheduled ticks finish with a persisted decision or explicit blocked reason over 24 hours.	Required
Snapshot freshness	At least 99% required-data coverage; alert after 60 seconds of continuous staleness.	Required
Decision latency	p95 under 3 seconds and always below the five-second decision budget; timeout becomes HOLD.	Required
Duplicate execution	Zero duplicate fills for one idempotency key.	Required
Ledger integrity	Equity, fills, funding, fees, and position reconcile on every restart and nightly.	Required
Risk-policy drift	Runtime hard-risk hash must equal deployment manifest; any mismatch halts new entries.	Required
Drawdown/liquidation safety	8% drawdown and 10% emergency liquidation-buffer paths flatten and halt in fault-injection tests.	Required
Notifications	Critical alerts delivered or visibly queued; duplicates suppressed.	Recommended
Cost telemetry	100% model calls record model, latency, token usage, and estimated cost.	Required

Health and recovery

- Docker health checks cover scheduler progress, stream freshness, database writes, provider reachability, and outbox backlog.
- The worker restarts automatically but becomes ready only after ledger reconciliation and a healthy feature snapshot.
- A nightly consistent SQLite backup is copied off-volume; seven daily and four weekly copies are retained.
- Telegram sends executed fills, promotion/rollback, kill-switch, sustained dependency incidents, and a daily summary at 09:00 Asia/Ho_Chi_Minh. Raw five-second signals are visible in the dashboard but are not individually pushed.
- Model spend is measured rather than assumed. The dashboard reports requests, tokens, latency, errors, and estimated daily cost per model/provider.

Verification and launch gates

Before a paper campaign starts, automated tests must cover:

- Closed-candle indicators and no-look-ahead replay.
- Feature synchronization, staleness, stream gaps, and REST rate limits.
- Valid, low-confidence, malformed, timed-out, and version-mismatched Jev responses.

- Position sizing under rounding, minimum quantity, two tranches, and both directions.
- Fees, funding, slippage, mark-to-market, maintenance margin, and liquidation price.
- Daily stop, drawdown kill-switch, liquidation-buffer reduction, and manual reset.
- Worker restart, database rollback, duplicate ticks, outbox retry, and rule promotion.
- Hyperliquid reconnect, frame checksums, cross-venue normalization, shadow isolation, and baseline-vs-overlay replay using identical decisions.
- Prompt-injection attempts in every external-text field.
- Dashboard secret redaction, authorization, CSRF, and command auditing.

The final pre-campaign gate is a 72-hour soak with zero duplicate fills, zero ledger reconciliation errors, no lost decision events, and successful fault injection for data/provider outages.

10. Alternatives Considered

Alternative	Why considered	Why not selected
Let LLM update rules_current.json directly	Fast adaptation and minimal implementation.	Breaks replay, allows mid-tick changes, and grants unvalidated text authority over trading behavior.
Select challenger after 30 minutes	Quick feedback.	Too few independent trades and too sensitive to one short market regime.
Poll every source through REST every five seconds	Simple implementation through one CCXT interface.	Produces mismatched timestamps, higher latency, and avoidable rate-limit pressure.
Use Jev only as advisory telemetry	Lowest model risk.	Does not test the intended product hypothesis that a typed System One model can own directional decisions inside deterministic safeguards.
Give Jev direct execution authority	Short hot path.	Model/provider errors could bypass risk, sizing, idempotency, and liquidation controls.
Use spot BTC/USDT	Simpler accounting and no liquidation.	Cannot faithfully exercise symmetric long/short intraday decisions or funding-related signals.
Use cross margin or hedge mode	More capital flexibility.	Enlarges blast radius and complicates attribution; isolated one-way state is safer and easier to replay.
Use PostgreSQL immediately	Strong multi-writer concurrency and mature operations.	A single-host, single-writer paper system does not yet justify the operational cost; contracts keep a future migration possible.

11. Open Validation Questions

No question blocks implementation of the paper design. The following hypotheses must be measured and must not be converted into assumptions:

1. Whether Jev direction confidence and toxic-flow scores are calibrated for this market domain. Store enough outcomes to create reliability diagrams by regime.
2. Whether five-second decisions improve net results after spread, slippage, funding, and churn compared with slower baselines.
3. Actual daily Jev and LLM cost using observed input sizes and call frequency.
4. Availability and quality of optional Coinglass/X feeds. The system must remain valid when both are disabled.

12. Decision and Next Steps

The recommended decision is to build this as a standalone, paper-only research system with Jev as the directional decision source and a deterministic risk kernel as the final authority. Isolated leverage is fixed at 3x, but the position is

capped at 50% notional, 20% initial margin, and 0.50% total stop risk. The current Donchian campaign and its data, account, scheduler, and promotion status remain untouched.

Milestone	Deliverable	Exit criteria
M1 — Deterministic foundation	Market-data recorder, synchronized snapshots, paper perpetual ledger, margin/risk engine, and replay fixtures.	Ledger reconciles; sizing and all kill switches pass unit/property tests without any model dependency.
M1.1 — Cross-venue shadow	Hyperliquid hybrid collector, normalized venue frames, shadow overlay, dashboard health, paired replay, and immutable promotion gate.	14 days, ≥95% coverage, ≥100 non-Hold decisions, ≥30 closed trades, and non-inferior risk metrics.
M2 — Jev shadow decisions	OpenRouter adapter, typed decisions, gate records, cost telemetry, and dashboard views; no paper fills initially.	72-hour decision soak meets freshness, latency, idempotency, and fail-closed gates.
M3 — Paper execution	Authorized simulated fills, Telegram operations, one champion, and one challenger ledger.	30 days without duplicate fills or reconciliation errors; risk interventions verified.
M4 — Adaptive rules	Hourly analysis, bounded candidates, 90-day replay, 14-day/30-trade evaluation, atomic promotion and rollback.	At least one full challenger evaluation is reproducible from stored inputs and passes promotion-policy tests.

Any proposal for live trading starts a separate architecture and security review. Paper performance alone is not approval to add exchange trading permissions.

References

- [TypeSafe System One concepts](#)
- [OpenRouter TypeSafe/Jev models](#)
- [OpenRouter API quickstart](#)
- [CCXT manual](#)
- [Binance derivatives API catalog](#)